Problem Set 6

Problem 1. (Level curves and tangent line). Find equations of the tangent line to the given level curves at the specified point.

(*i*) $x^2 + y^2 = k^2$, $(k, 0)$

(*ii*) $xy = k$, $(1, k)$

(*iii*) $y - x^2 = k$, $(1, k + 1)$

Problem 2. (Level surfaces and tangent plane). Find equations of the tangent plane to the given surface at the specified point.

(*i*) $z = x^2 + y^2$, $(1, 1, 2)$

(*ii*) $y = x^2 - z^2$, $(4, 7, 2)$

(*iii*) $xy + yz + zx = 5$, $(1, 2, 1)$

(*iv*) $\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$, $(x_0,\ y_0,\ z_0)$

Problem 3. (Chain rule) Use chain rule to find $\frac{df}{dt}$ of the following functions

(*i*) $f(x, y) = \frac{x^3+y^3}{3}$, $x = 4\ \cos t$, $y = 3\ \sin t$

(*ii*) $f(x, y) = x^2 + y^2 + xy$, $x = \ \sin t$, $y = e^t$

(*iii*) $f(x, y) = \sqrt{1 + x^2 + y^2}$, $x = \ln t$, $y = \ \cos t$

(*iv*) $f(x, y, z) = \ \ln\sqrt{x^2 + y^2 + z^2}$, $x = \ \sin t$, $y = \ \cos t$, $z = \ \tan t$

Problem 4. (Implicit differentiation)

(*i*) Find the implicit derivative $\frac{dy}{dx}$ of the following implicit functions

(a) $x^2 - 5xy + 2y^2 = 1$

(b) $x^3 + y^3 - 6xy = 0$

(c) $y\cos x = x^2$

(d) $y^2 = \cos(xy) = 1 + \ \sin y$

(*ii*) Find the implicit partial derivative $\frac{\partial z}{\partial x}$, $\frac{\partial z}{\partial y}$ of the following implicit functions

(a) $x^2 + 2y^2 + 3z^2 = 1$

(b) $x^2 - y^2 + z^2 - 2z = 4$

(c) $x^3 + y^3 + z^3 + 4xyz = 1$

(d) $yz + x\ln y = z^2$

Problem 5. (Mean value theorem)

(i) Let $\mathbf{x_0} \in \mathbb{R}^n$. Show that the set $S = \{\mathbf{x} : ||\mathbf{x} - \mathbf{x}_0|| < r\} \subset \mathbb{R}^n$ is a convex set.

(ii) Let $f : \mathbb{R}^n \to \mathbb{R}$ be any function. Assume that $D_{\mathbf{u}}f(\mathbf{x}) = 0$ for all $\mathbf{x}$ in some open convex set S and for all unit vector $\mathbf{u} \in \mathbb{R}^n$. Show that f is constant on S

(iii) Let f denotes the differentiable function defined on an open convex set S. Then prove that $\nabla f = 0$ iff f is constant on S.

Problem 6. (Linear map).

(i) Write a linear map which rotate all vectors in $\mathbb{R}^2$ anticlockwise by angle θ.

(ii) If a mirror placed along a line making an angle θ. What is the formula for reflection about a line.

(iii) Given a line passing through origin, find the formula for projection onto that line.

(iv) Find the linear map which maps the points in three dimensional space $\mathbb{R}^3$ onto the points in the xy-plane.

(v) Given a linear map $A : \mathbb{R}^n \to \mathbb{R}^m$ there exist a $m \times n$ matrix $\tilde{A}$ such that $A(\mathbf{x}) = \tilde{A}\begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}$ and vice versa.

Problem 7. Find the 2nd degree Taylor polynomial at those points where the gradient is $\mathbf{0}$:

(i) $f(x, y) = xy^2 + x^2y$

(ii) $f(x, y) = x^2 + (y - 1)^2$

(iii) $f(x, y) = 1 + x^2 - (y - 1)^2$

(iv) $f(x, y) = (x - y + 1)^2$

(v) $f(x, y) = \sin x \sin y \sin(x + y)$ where $x, y \in [0, \pi]$

(vi) $f(x, y) = x^2 - xy + y^2 - 2x + y$

(vii) $f(x, y) = 2x^2 - xy - 3y^2 - 3x + 7y$

(viii) $f(x, y) = x^3 - 3xy^2 + y^3$

(ix) $f(x, y) = x^2y^3(6 - x - y)$

(x) $f(x, y) = e^{2x+3y}(8x^2 - 6xy + 3y^2)$

Problem 8. Find the Hessian of the above functions at all those points where the gradient is $\mathbf{0}$.

Problem 9. Find the first and second degree Taylor polynomials of the function $f(x, y) = e^x \cos y$ at origin.